

Sentiment Classification for Kinyarwanda: A Comparison of Clean and Code-Mixed Multilingual Pretraining

NLP Summative Project — Text Classification for an African Language

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GitHub Repository: github.com/Iyamurinze/summative

Model (Hugging Face Hub): huggingface.co/Iyamurinze/afroxlmr-kinyarwanda-sentiment

Live System: [\[INSERT DEPLOYED STREAMLIT URL\]](#)

Demo Video: [\[INSERT DEMO VIDEO LINK\]](#)

1. Abstract

This project develops and deploys a sentiment classifier for Kinyarwanda, a Bantu language spoken by over 12 million people primarily in Rwanda, using the AfriSenti Twitter sentiment dataset (Muhammad et al., 2023). A TF-IDF with classical machine learning baseline (Logistic Regression and Linear SVM) is compared against two fine-tuned Transformer models: AfroXLMR (Alabi et al., 2022), an XLM-R checkpoint adapted on 17 African languages including standalone Kinyarwanda, and AfriBERTa (Ogueji et al., 2021), a model pretrained from scratch on 11 African languages whose only Kinyarwanda-adjacent exposure is "Gahuza," a code-mixed Kinyarwanda/Kirundi corpus. On a held-out test set of 1,026 tweets, the Logistic Regression baseline reaches 58.4% macro-F1, AfriBERTa reaches 63.9%, and AfroXLMR reaches 67.5%, a +9-point improvement over the baseline. AfroXLMR's consistent outperformance of AfriBERTa across all three sentiment classes supports the hypothesis that clean, language-specific pretraining data transfers better to a downstream task than code-mixed exposure to a related language, even when the two languages are typologically close. Error analysis shows the models handle explicit, lexically-marked sentiment well but struggle with implicit or rhetorical sentiment expressed through tone and cultural context rather than vocabulary. The final model is deployed through a Streamlit web application backed by the fine-tuned AfroXLMR checkpoint hosted on the Hugging Face Hub.

2. Introduction

Natural language processing research and tooling remain heavily skewed toward a small number of high-resource languages, leaving most African languages with limited labeled data, few pretrained models, and almost no deployed consumer-facing NLP applications (Muhammad et al., 2023). Sentiment analysis of social media text is one of the most immediately useful NLP applications for these languages: it enables monitoring of public opinion, customer feedback, and public communication at a scale manual reading cannot match. This project builds and deploys a sentiment classifier for Kinyarwanda social media text, assigning each input a label of `positive`, `negative`, or `neutral`.

Kinyarwanda was selected for two reasons. First, it is a genuinely under-resourced language relative to its speaker population of over 12 million, with comparatively few deployed sentiment or classification tools. Second, and more directly motivating the final design of this project, the author is a Kinyarwanda speaker. An earlier iteration of this project used Hausa, for which a strong dataset and pretrained models are also available; however, the author does not read Hausa, which made genuine qualitative error analysis, honest interpretation of model mistakes, and a defensible oral explanation of the system's behavior impractical. The project was deliberately rebuilt around Kinyarwanda so that every reported example, error, and design decision could be verified and explained first-hand rather than taken on faith from automated metrics alone. This is a practical, not merely academic, point: a central requirement of this coursework is the ability to defend the submitted work, which requires the author to actually understand the language the system operates on.

The objective of this project is threefold: (1) establish a classical machine learning baseline for Kinyarwanda sentiment classification, (2) fine-tune and compare two pretrained Transformer models that differ specifically in whether their pretraining data included clean, standalone Kinyarwanda or only code-mixed Kinyarwanda/Kirundi text, and (3) deploy the best-performing model through a public web interface.

3. Related Work

AfriSenti (Muhammad et al., 2023) is, to date, the largest sentiment analysis benchmark for African languages, covering 110,000+ annotated tweets across 14 languages including Kinyarwanda, released as part of the SemEval-2023 Task 12 shared task. It provides the labeled data this project relies on and is the only resource of its kind offering naturally-occurring, human-annotated Kinyarwanda sentiment data at this scale.

Two lines of prior work on pretrained Transformers for African languages motivate the model comparison in this project. Alabi et al. (2022) adapt XLM-R (Conneau et al., 2020) to 17 African languages through continued masked-language-model pretraining, producing AfroXLMR; Kinyarwanda is one of the 17 languages included, pretrained as a standalone language distinct from Kirundi. Ogueji et al. (2021) take a different approach with AfriBERTa,

pretraining a smaller Transformer from scratch on a corpus of only 11 African languages with no general multilingual or English pretraining at all; for Kinyarwanda specifically, AfriBERTa's only exposure is through "Gahuza," a code-mixed Kinyarwanda/Kirundi social media corpus, not pure Kinyarwanda, a fact confirmed directly from the model's official documentation on the Hugging Face Hub. No prior published work, to the author's knowledge, directly compares these two specific pretraining strategies on a Kinyarwanda-only downstream task, which is the gap this project's experiment addresses.

Beyond sentiment, Niyongabo et al. (2020) introduce KINNEWS and KIRNEWS, news topic classification datasets for Kinyarwanda and Kirundi, demonstrating that labeled text classification resources for Kinyarwanda exist but remain limited relative to higher-resource languages. This project's approach — comparing a classical TF-IDF baseline against fine-tuned Transformers — follows the standard methodology established across this body of work for low-resource African language classification tasks.

4. Dataset and Data Preparation

The dataset is the Kinyarwanda subset of AfriSenti-Twitter (Muhammad et al., 2023), accessed via the script-free parquet mirror `mteb/AfriSentiClassification` on the Hugging Face Hub, since the original `HausaNLP/AfriSenti-Twitter` repository ships a legacy loading script incompatible with current versions of the `datasets` library. The mirror preserves the official AfriSenti splits and labels.

The dataset contains 3,302 training, 827 validation, and 1,026 test tweets. Labels are `positive`, `negative`, and `neutral`, and all three splits are reasonably class-balanced: the training set contains 1,257 negative, 1,146 neutral, and 899 positive tweets; the test set contains 393 negative, 355 neutral, and 278 positive tweets. This is a meaningfully smaller dataset than some other AfriSenti languages — approximately 4x smaller than the Hausa subset of the same dataset (14,172 training tweets) — which is discussed further as a limitation in Section 9.

The data consists of real, naturally-occurring, human-annotated tweets, not synthetic or machine-generated text, satisfying the project's requirement to work with authentic, naturally-occurring data. As Twitter text, it contains `@mentions`, URLs, emojis, and informal spelling typical of social media. During preprocessing, `@mentions` and URLs are normalized to placeholder tokens (`<USER>`, `<URL>`) since they carry no sentiment signal and would otherwise inflate vocabulary size for the TF-IDF baseline. Emojis and punctuation are deliberately preserved, since both frequently carry genuine sentiment signal in tweets (e.g. 🤔, ❤️, repeated exclamation marks). No additional train/validation/test splitting was performed, as AfriSenti provides official splits as part of the SemEval-2023 shared task.

5. Methodology

5.1 Baseline

The baseline represents each tweet as a TF-IDF vector over unigrams and bigrams, capped at 20,000 features with a minimum document frequency of 2, then classifies with Logistic Regression and, as a second experiment, a Linear Support Vector Machine, both using balanced class weights to account for any residual class imbalance. This baseline establishes the performance achievable without any contextual language understanding, purely from surface-level word and bigram frequency patterns — the number the fine-tuned Transformers must beat to justify their additional complexity and compute cost.

5.2 Proposed Models

Two pretrained Transformer checkpoints are fine-tuned identically, each with a 3-class classification head added on top of the base encoder, trained end-to-end:

- **AfroXLMR** (`Davlan/afro-xlmr-base`; Alabi et al., 2022) — an XLM-R-base checkpoint continually pretrained on 17 African languages plus Arabic, French, and English, with Kinyarwanda included as a standalone language.
- **AfriBERTa** (`castorini/afriberta_large`; Ogueji et al., 2021) — a 126M-parameter Transformer pretrained from scratch on 11 African languages; its only Kinyarwanda-adjacent pretraining data is "Gahuza," a code-mixed Kinyarwanda/Kirundi corpus.

Both models are fine-tuned with identical hyperparameters to isolate the effect of the pretrained checkpoint: learning rate 2e-5, per-device batch size 16, 3 epochs, weight decay 0.01, maximum sequence length 64 tokens, and the best checkpoint selected by validation macro-F1 (`metric_for_best_model="f1"`) using the Hugging Face `Trainer` API. This comparison directly tests whether exposure to clean, standalone Kinyarwanda during pretraining (AfroXLMR) transfers better to a pure-Kinyarwanda downstream task than exposure to code-mixed Kinyarwanda/Kirundi text (AfriBERTa), even though Kinyarwanda and Kirundi are closely related Bantu languages.

5.3 Evaluation Protocol

The baseline models are evaluated on the validation split. The fine-tuned Transformers are evaluated on the full test set by loading the exact checkpoints pushed to the Hugging Face Hub and running inference directly — ensuring the reported numbers reflect the actual deployed artifact, not an in-session training metric that could differ from what end users interact with.

6. Experiments and Results

Table 1 summarizes all four approaches. Accuracy and macro-averaged Precision/Recall/F1 are reported; macro-F1 is treated as the primary metric since it weighs all three sentiment classes equally regardless of their (near-)balance, which is appropriate given the classification task has no class of overriding importance.

Approach	Split	Accuracy	Macro-F1
TF-IDF + Logistic Regression	Validation	58.16%	58.38%
TF-IDF + Linear SVM	Validation	56.47%	56.66%
Fine-tuned AfriBERTa	Test	63.65%	63.88%
Fine-tuned AfroXLMR	Test	67.15–67.93%	67.47–68.28%

Table 1. Comparison of all four approaches. AfroXLMR was trained twice to check reproducibility; both runs are reported as a range (67.15–67.93% accuracy, 67.47–68.28% macro-F1) given the expected run-to-run variance from not fixing a random seed. Both runs beat AfriBERTa by a consistent margin.

Both fine-tuned Transformers substantially outperform the classical baseline. AfroXLMR improves macro-F1 by approximately 9 points over the strongest baseline (Logistic Regression), compared to only a ~4-point improvement the same pipeline achieved on the higher-resource Hausa subset of this same dataset in an earlier iteration of this project. This larger gap in a lower-resource setting is itself an interesting finding: with less training data, the bag-of-words baseline has proportionally less signal to learn from, while the pretrained Transformer’s prior language knowledge compensates more — pretraining appears to matter *more*, not less, as available task-specific data shrinks.

More directly addressing the experiment’s central question, AfroXLMR consistently outperforms AfriBERTa by roughly 3.6–4.4 points macro-F1 across two independent training runs. Table 2 breaks this down by class for a single representative run of each model.

Class	AfriBERTa F1	AfroXLMR F1
Negative	0.619	0.651
Neutral	0.637	0.685
Positive	0.661	0.713

Table 2. Per-class F1 on the test set. AfroXLMR outperforms AfriBERTa in every individual class, not merely on the aggregate macro-F1 — evidence that the improvement reflects a genuinely better internal representation of Kinyarwanda, not an artifact of class weighting.

This per-class consistency is important: because AfroXLMR wins on all three classes rather than trading off strength in one class for weakness in another, the aggregate macro-F1

improvement is not an artifact of how the macro-average happens to weight the classes. This supports interpreting the result as a genuine representational advantage from AfroXLMR's exposure to clean Kinyarwanda during pretraining, directly confirming the hypothesis set out in Section 5.2.

7. Error Analysis and Discussion

Across both Transformer models, the `negative` class consistently shows the lowest per-class F1, and recall for `negative` is the lowest of the three classes for AfroXLMR (0.649 vs. 0.693 for neutral and 0.705 for positive), indicating the model under-identifies negative sentiment relative to the other two classes more often than it under-identifies positive or neutral sentiment.

Qualitative inspection of individual predictions clarifies why. Correctly classified examples tend to express sentiment explicitly and lexically — for instance, *"Mwese muzagire Umwaka mushya muhire wa #2022"* ("May you all have a happy new year 2022") is correctly classified positive with 94% confidence, and *"Ibintu 3 byo kureba: Intumbero, Imyitwarire, Ururimi."* is correctly classified negative with 77% confidence. By contrast, a representative misclassification is *"Ngo iyo Imana iza kwica satani, satani yari kuba yitabye Imana #fact"* (roughly, "They say if God were to kill Satan, Satan would have already surrendered to God"), labeled negative in the dataset but predicted neutral with 68% confidence. This tweet expresses its sentiment through a philosophical/religious rhetorical structure rather than explicit negative vocabulary — the kind of implicit, culturally-situated sentiment that is inherently harder for a model to learn from a training set of only 3,302 examples, since such rhetorical patterns appear too infrequently for the model to generalize a reliable signal from them.

This pattern is consistent with the broader literature on pretraining-data relevance: Alabi et al. (2022) find that target-language-adapted pretraining consistently improves downstream task performance over generic multilingual pretraining, which this project's AfroXLMR-vs-AfriBERTa comparison independently reproduces at a finer grain — not just "more adaptation helps," but specifically that clean exposure to the exact target language outperforms exposure to a closely related but distinct language (Kirundi, via Gahuza), even though the two languages share substantial vocabulary and grammar as neighboring Bantu languages. The remaining error patterns — difficulty with implicit, rhetorical, or culturally-coded sentiment — are a function of limited training data rather than a flaw specific to either pretrained checkpoint, and would likely improve with a larger labeled dataset or techniques such as cross-lingual transfer from higher-resource AfriSenti languages before fine-tuning on Kinyarwanda specifically (see Section 9).

8. Deployment

The fine-tuned AfroXLMR model is hosted on the Hugging Face Hub at `Iyamurinze/afroxlmr-kinyarwanda-sentiment`, from which it is loaded directly by a Streamlit web application (`app/streamlit_app.py` in the project repository). The application exposes a simple text box where a user enters Kinyarwanda text and receives the predicted sentiment label along with the model's confidence for all three classes, plus a set of pre-loaded example tweets (with known gold labels) demonstrating both correct predictions and a genuine misclassification. The app is deployed on Streamlit Community Cloud, which was chosen specifically because it requires no payment or card details for public apps, after Hugging Face Spaces' Gradio/Docker SDKs were found to require a paid plan at the time of this project. A parallel Gradio implementation (`app/app.py`) is included in the repository for local use. Both applications load the identical pushed model, so their predictions are equivalent.

9. Limitations and Future Work

- **Dataset size.** At 3,302 training tweets, the Kinyarwanda subset of AfriSenti is roughly 4x smaller than some other AfriSenti languages, which likely constrains how well either fine-tuned model can learn, particularly for implicit or rhetorical sentiment patterns that require more examples to generalize from.
- **Single-run training.** Models were trained without a fixed random seed; reported AfroXLMR results vary by roughly 0.4–0.8 points macro-F1 across two independent runs. A more rigorous evaluation would average results across multiple seeds.
- **Domain narrowness.** Both models are trained exclusively on Twitter data; performance on other Kinyarwanda text domains (news, formal writing, customer service transcripts) is untested and likely to differ.
- **No hyperparameter search.** Learning rate, batch size, and epoch count were fixed by convention rather than tuned; a learning-rate sweep or longer training with early stopping may yield modest further gains.
- **Future work** could explore cross-lingual transfer — fine-tuning first on the full multilingual AfriSenti training set before specializing on Kinyarwanda — to partially compensate for the limited Kinyarwanda-specific data, as well as ensembling AfroXLMR and AfriBERTa predictions, or evaluating a larger AfroXLMR variant.

10. Conclusion

This project built and deployed a Kinyarwanda sentiment classifier, establishing a TF-IDF classical baseline and comparing two fine-tuned Transformer models that differ specifically in whether their pretraining data included clean, standalone Kinyarwanda or only code-mixed Kinyarwanda/Kirundi text. Both Transformers substantially outperformed the baseline, with

AfroXLMR reaching 67.5% macro-F1 against the baseline's 58.4% and AfriBERTa's 63.9%. AfroXLMR's consistent per-class advantage over AfriBERTa provides direct empirical evidence that clean, language-specific pretraining data transfers better than code-mixed exposure to a closely related language, even between typologically similar Bantu languages. Error analysis showed that both models handle explicit, lexically-marked sentiment reliably but struggle with implicit, rhetorical sentiment — a limitation plausibly tied to the dataset's limited size rather than to either pretrained checkpoint specifically. Beyond the technical result, choosing a language the author could genuinely read and verify proved essential to producing honest, defensible error analysis rather than metrics taken on faith — a lesson as relevant to doing this kind of project well as any individual modeling choice.

11. References

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